

*All data and figures are sourced from [1]

Figure A:

This figure illustrates the effect of TGF β with or without Y-27632 on the transformation of lens epithelial cells (LEC) into mesenchymal cells. The column labels indicate which substance is being 'colored' in each column. F-actin and α -SMA are indicators of mesenchymal cells; the Merge+DAPI column overlays the other two columns, and stains living cells with blue. The row label is the treatment applied to panels in that row. The control row is untreated. The horizontal bar in the top-left panel indicates the scale of the panels.

Comparing the first and third rows supports the hypothesis that TGF β induces the formation of mesenchymal cells. The third row has no red/green, indicating no mesenchymal cells are usually present. In the first row, the panel is covered in red and green, suggesting that treatment with TGF β is what induced transformation into mesenchymal cells.

Additionally, the second row (in comparison to the other two), supports the hypothesis that Y-27632 counteracts TGF β 's theorized effect of transforming LEC into mesenchymal cells. If the first row implies TGF β has a role in LEC transformation, then the second row may indicate that Y-276232 counteracts it, as despite its treatment it is very similar to the control row.

The third column further suggests TGF β facilitates LEC transformation, as the nuclei (blue portions) in the top panel are larger and sparser than the other two panels in the column, suggesting a structural transformation.

Figure B:

This figure visualizes the relative quantities of proteins under certain treatments. The top 2 labels indicate treatment, with +/- signalling treatment presence/absence. The bottom 2 labels display relative quantities of their respective proteins in each treatment condition. The third lane is a control that displays standard α SMA levels, while the fourth lane is a treated control that displays the effect of the inhibitor Y-27632 alone. The GAPDH indicates the quantity loaded onto each lane, ensuring that lanes are comparable.

Lane 1 in comparison to lane 3 suggests that TGF β leads to a higher-than-baseline amount of α SMA, and the two are comparable because their GAPDH blots are similar in intensity. Subsequently, lane 2 suggests that Y-27632 counteracts the effect of TGF β as it has a much lighter α SMA band. However, this counteractive effect is likely not complete since lane 2

still has a slight α SMA band compared to lane 3. Lane 4 suggests Y-27632 does not stimulate the quantity of α SMA. All of this supports the hypotheses about TGF β and Y-27632.

Figure C:

The x-axis indicates the groups with and without TGF β , while the y-axis indicates the ratio of α SMA to GAPDH; since all cells have GAPDH, this denotes the proportion of cells that have α SMA, allowing for comparison between treatments. Black bars are also treated with Y-27632; white bars are not. The smaller horizontal line indicates that the two treatments with TGF β are statistically significant ($p \leq 0.01$), while the larger one indicates that treatments without Y-27632 are statistically significant ($p \leq 0.01$).

Comparison of the first bar to the third suggests that TGF β treatment led to more cells with α SMA, since the y-value for bar 1 is greater than bar 3, which is the basal amount. Then, comparing bar 1 to bar 2 suggests that Y-27632 reduces this effect of TGF β , potentially counteracting it entirely as the quantity of α SMA in bars 2 and 3 have overlapping error bars. Once again, both hypotheses are supported by these data.

There is no statistically significant effect due to treatment with just Y-27632, indicated by the lack of a horizontal line between bar 3 (control) and bar 4 (just Y-27632).

References

1. Korol A, Taiyab A, West-Mays JA. RhoA/ROCK signaling regulates TGF β -induced epithelial-mesenchymal transition of lens epithelial cells through MRTF-A. (1528-3658 (Electronic)).